Census of 321 Tires — Real vs Nominal Width

Constant-deflection model · BikeLab PSI

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Abstract

Census of 321 road and gravel tires recording measured real width versus nominal, corrected for internal rim width. It is the empirical core of the constant-deflection model (Berto 15% criterion) of the BikeLab PSI Calculator, cross-validated against SILCA and Rene Herse (9/12 wheels within ±10%). Uncertainty via Latin Hypercube (10,000 samples) and Sobol indices.

Verifiable findings

Road: tires measure ~+0.8 mm over nominal on an 18 mm internal rim. Gravel: ~-1.8 mm under nominal. Sobol: on road real width dominates (S1 up to 0.68); on gravel caliper gauge error dominates (S1 up to 0.73). Rider weight explains under 9%. ETRTO hookless caps: 72.5 PSI (60 PSI at 30 mm internal or more).

Sample

Case	Configuration	Mean [p5-p95]	Band +/-PSI
1	83 kg · 28 mm · ruta · aro 21	73 [63-86]	11,7
2	95 kg · 28 mm · ruta · aro 21	84 [72-100]	13,8
3	75 kg · 32 mm · ruta · aro 19	63 [54-73]	9,2
4	85 kg · 40 mm · gravel · aro 24	38 [35-41]	3,2
5	100 kg · 45 mm · gravel · aro 25	39 [36-41]	2,5
6	70 kg · 25 mm · ruta · aro 21	71 [59-85]	12,8

Dataset variables

nominal width (mm); measured real width (mm); internal rim width (mm); recommended pressure (PSI); uncertainty band (±PSI). The full set (321 entries), correction factors and sampling code are shared under agreement.

How to cite:

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Provisional by design. A verifiable sample is published; the full raw set and the reproducible method are shared under agreement. Contact: carlos.ravello@zoovettravel.com